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The Deformation and Ageing of Mild Steel: III Discussion of Results

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ABSTRACT An attempt is made here to explain the observed phenomena in the yielding and ageing of mild steel, described in two previous papers, in the general terms of a grain-boundary theory. On this hypothesis, a satisfactory explanation of the variation of the lower yield point with grain size may be developed. It is shown that strain-ageing must involve two processes: a healing of the grain-boundary films, coupled with a hardening in the grains themselves. A discussion of the possible nature of the grain-boundary film is also undertaken.

§ 1. INTRODUCTION

IN two previous papers (Sylwestrowicz and Hall 1951, Hall 1951, which will be referred to as I and II) an experimental study has been made of some of the factors which influence the behaviour of mild steel in tensile tests. In this paper, a little more experimental evidence will be supplied, and the results correlated qualitatively in terms of a grain-boundary theory.

§ 2. THE UPPER YIELD POINT

There is now no doubt, from the work of Fast (1950) and others, that small amounts of carbon and nitrogen present in a matrix of ferrite will cause the yield point and ageing phenomena found in mild steel. In I, a comparison was made between the two current theories of the upper yield point, the old grain-boundary theory of Nadai (1924) and Kuroda (1938), and the more recent 'dislocation clogging' mechanism of Cottrell and Bilby (1949). Without identifying any particular component at the grain-boundary, it was suggested in that paper that some form of grain-boundary theory was more likely to explain the observed results.

The one objection to the grain-boundary theory most frequently heard (Cottrell 1948, Bilby 1950) is that once the matrix of ferrite has yielded, the grain boundary films alone have to withstand the load. Arguing from an analogy of a series of hard, brittle wires and soft ductile ones in parallel array pulled together in tension, Bilby (1950) has shown that a model with correct proportions of these brittle and plastic elements would have an extremely small upper yield point. In the opinion of the present writer, however, this analogy is incorrect, and it would be wise to examine the 'honeycomb' hypothesis of Kuroda in the light of our present knowledge of dislocation theory.

It is known that if a dislocation in a matrix approaches a region, such as a grain-boundary film, which has a higher yield stress than the matrix itself, then this region constitutes an effective potential barrier for the passage of the dislocation. In other words, the dislocation will experience a repulsion near the grain boundary interface, and dislocations will pile up along the glide plane behind the grain boundary film, until the stress concentrations around the tip of the slip band cause the film to yield. Deformation is then transferred to the next grain.

Cottrell (1949) has shown that the force on such an obstacle from an array of n dislocations of like sign is given by $P = na\sigma$ where a is the interatomic spacing, and σ the applied shear stress, or, in other words, the dislocation array produces a local stress concentration in the vicinity of the obstacle of n times the external stress.

This stress concentration which the boundary must withstand is very much less than in Bilby's model, where it is assumed that all the load is being carried by the thin boundary films. A direct comparison of the two models may be made if we let L be the length of the slip plane, l the spacing of the dislocations, and b the thickness of the boundary film. Then if after the ferrite yields, it carries no stress, the stress concentration factor in the film is L/b . In the model based on an array of dislocations, the stress concentration is n where $nl = L$. For a grain diameter of 10^{-3} cm. and a maximum thickness for the film of ten atomic spacings, $L/b \approx 10^4$. In the other case, we must assume a dislocation density of 10^8 dislocation lines/cm² in the annealed ferrite, or an average of ten dislocation lines per slip plane in material of the same grain size. Provided no multiplication of the dislocations takes place until the boundary film yields, this model gives a stress concentration factor about a thousand times smaller than in the model composed of ductile and brittle wires, and a more reasonable value for the physical strength of the boundary.

The exact nature or the strength of the boundary films must still remain unknown; some of its necessary properties will be examined in the final paragraph of this paper. The concept of dislocation arrays will be extended further in the next section on the lower yield point.

In the first paper in this series, it has been suggested that a theory of this nature would be necessary to explain the observed dependence of the upper yield point on grain size. It is interesting to see how this theory, coupled with the 'nucleation' hypothesis advanced in the first paper, gives the correct qualitative grain-size dependence. The larger the grains, the more rapidly is the size of the plastically deformed nucleus increased by the breakdown of a single boundary, and the more rapidly is the state reached when the nucleus will spread to form the first Lüders band. Thus, one would expect the difference $\sigma_{\text{UYP}} - \sigma_{\text{LYP}}$ to be smaller the larger the crystal size, and the results in I (Table 2) confirm this.

§ 3. THE LOWER YIELD POINT

If this concept of the grain-boundary theory is correct, then it is possible to explain many of the facts about the lower yield point, and in particular, its variation with grain size.

The relationship between the lower yield point and grain size has been determined experimentally by the writer, using the three materials referred to in I, and the results are displayed in Figure 1. It is seen that all the points, with the exception of the coarsest-grained Bessemer steel, lie on a single curve. The Bessemer steel, however, contains a much larger amount of nitrogen than the other two, and a high level of impurity, but it is interesting that the Armco and open-hearth steels, containing 0.02% and 0.06% carbon respectively, have identical values of the lower yield point within the limits of experimental error. The full curve in this Figure lies parallel to and a little below the curve of results of Winlock and Leiter (1937); their results, however, covered a smaller range of grain sizes, and were taken at much higher strain rates (0.002/min., instead of 10^{-4} /min. as in the present case). Thus, higher values of the lower yield point would be expected in their case.

If instead of Figure 1, values of σ_{LYP} from the full curve are plotted against the inverse square root of the grain diameter d , it is found that the relation between the two quantities is linear (Figure 2); the graph cuts the y axis at the

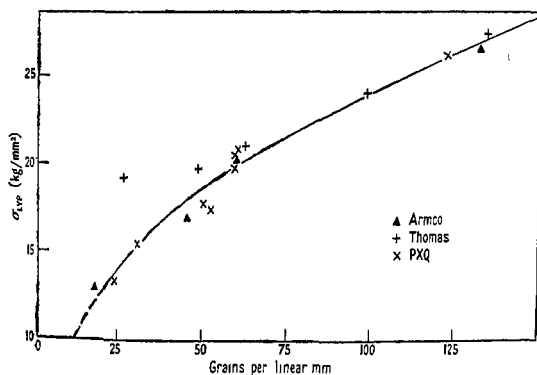


Figure 1.

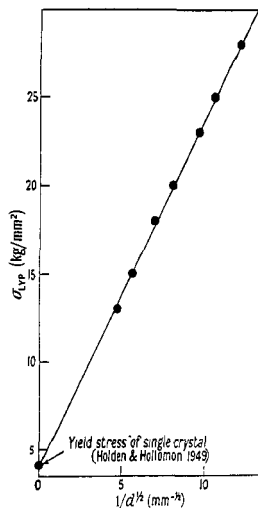


Figure 2.

value for the yield stress of single crystals determined by Holden and Hollomon (1949a). Thus

$$\sigma_{LYP} - \sigma' \propto 1/d^{1/2} \quad \dots\dots (1)$$

where σ' is the yield stress for a single crystal.

This relationship follows from theoretical calculations by Eshelby, Frank and Nabarro (1951). The model used in their work is outlined in the following paragraph.

The stress at a distance r from a dislocation is of the form A/r , where A is a constant. Let there be a succession of n of these dislocations of like sign, forced in the distance of increasing x by a uniform applied shear stress σ_0 , while

the leading dislocation is held at $x=0$ by an obstacle, such as a grain-boundary film, which exerts no long-range forces. Equilibrium is then established between applied force, the obstacle and the mutual repulsion of the dislocations. It may then be shown that, to a close approximation, the distance between the leading and the second dislocation is $d' \approx 2A/n\sigma_0$ while the distance between the first and last dislocation is $L \approx 2An/\sigma_0$.

The additional stress σ at a point on the glide plane a distance x ahead of the first dislocation, such that $d'/15 \ll x \ll L$ is given by

$$\sigma/\sigma_0 \approx (L/x)^{1/2}. \quad \dots\dots(2)$$

This relationship is exactly analogous to the stress concentration at a distance x from the end of an internal crack of the type postulated by Griffiths (1920) to explain the low brittle strength of solids.

If we suppose that the grain-boundary film will break down when the additional stress σ reaches a certain critical value, then the deformation will be transmitted to the next grain. This state will occur when the end dislocation is at a fixed distance x from the boundary: furthermore, we may identify L , the length of the slip band, with the mean grain diameter d . Thus, rearranging, we find that $\sigma_0 \propto 1/d^{1/2}$.

However, since flow is impossible in a single crystal at stresses below its yield stress σ' , then σ_0 in the analysis should be replaced by $\sigma_0 - \sigma'$ giving the grain-size relation as

$$\sigma_0 - \sigma' \propto 1/d^{1/2}, \quad \dots\dots(3)$$

which is of the same form as (1). This final step implies that each dislocation is subject to a uniform retarding force σ' in its passage through each grain. It should be mentioned here that this relationship between the lower yield stress and grain size may break down at grain diameters larger than those investigated in this work, for, as Morrison (1939) has shown, the yield stress of a thin-walled tube containing only a few grains in its cross section may be abnormally low.

The variation of the lower yield stress with strain rate is more difficult to deal with mathematically, although one can easily see why such an effect does exist. During the shear process at the front of the Luders band, described in the second paper in this series, the grain-boundary films are broken down, and flow then occurs by creep under the lower yield stress. Creep is a time dependent process, so that the dislocation arrays will be built up relatively slowly, and an appreciable interval will elapse before the next boundary film breaks down. As the strain rate increases, the dislocations must be speeded up to keep pace with the deformation, and this can only be done if the whole stress level rises at the same time. And as the propagation stress increases, the Luders strain also rises (see I).

The grain-boundary theory also explains the appearance of diffuse Luders bands in coarse-grained material (see II). Here the individual grains are so large that part of the front of deformation may get well ahead of another part which is held up, say, near a patch of smaller included grains. Then the sharp shear kink disappears, and we have the phenomenon of a typical diffuse band.

§ 4. STRAIN-AGEING PHENOMENA

When the Luders band covers the entire gauge length of the specimen, all the grain boundary films are broken down and weakened, and the material is able to deform homogeneously. During the ageing process, the films begin

to heal around the strain-hardened grains by the diffusion of carbon and nitrogen, with a resulting increase in their yield stress and the yield stress of the whole specimen, and deformation must again occur by a Lüders band—the secondary band. In lightly aged material, however, the boundaries have not regained their full strength, and the effect is the same as in material where the grain boundaries are far apart. Deformation is by a diffuse band. As the ageing becomes more advanced, more and more boundaries become restored to their original state, the front of the band is straightened and becomes sharp.

This analysis of course only applies to material which was sufficiently fine-grained to deform by a sharp band in the virgin state. If the grain size is so large that the initial deformation was by a diffuse band, then naturally no amount of straining and ageing will bring the grain boundaries any closer together, and deformation will always be by diffuse bands.

Thus, as the grain-boundary films reform around the strain-hardened material, the yield stress of the bulk material rises, and a set of curves similar to Figure 3 of I is obtained. However, it is not possible, from a set of curves such as this, to calculate the activation energy for the ageing process, because two processes are at work. In addition to the healing of the boundary films, Holden and Hollomon (1949b) and Cottrell and Churchman (1949) have also shown that single crystals may strain-age—consequently we must consider the individual grains of the matrix as hardening after the deformation as well. Each of these processes will raise the yield stress by different amounts, and may actually occur at different rates, for the diffusion constants of carbon and nitrogen in α -iron may be structure sensitive. Consequently, one cannot obtain the activation energy simply by measuring the time needed to reach a certain value of $\sigma_{\text{UYP}} - \sigma_{\text{LYP}}$ at various temperatures. The values obtained from the curves in I are all much lower than the accepted value of 19.8 kcal/mole (Wert and Zener 1949), but tend to this value at higher values of $\sigma_{\text{UYP}} - \sigma_{\text{LYP}}$. Nabarro (1948) was able to calculate this value for the activation energy from the hardness measurements of Davenport and Bain (1935) at a very late stage of ageing, where it seems that only one process is predominant. Thus, the activation energy determinations in this work do not prove that the atomic processes involved are other than the diffusion of carbon and nitrogen.

Another effect may result from this hardening of the individual grains of the matrix. The elongation to fracture of strain-aged specimens is very much less than the virgin material, and decreases the longer the ageing time. This lowering of the ductility is quite unexpected, and cannot be deduced from either the dislocation mechanism, or from the grain-boundary theory alone. Further work on the properties of strain-aged single crystals will be needed to explain this anomaly.

§ 5. THE NATURE OF THE GRAIN BOUNDARY FILM

It is not possible from this work to give any definite answer on the nature of the grain-boundary film, but in the conclusion to this paper it will be convenient to discuss some of its necessary properties.

In the first place, the yield strength of the grain-boundary film must not be too high, for in this case the grains cannot deform at all and the material becomes brittle. Examples are found in copper and gold, alloyed with small amounts of bismuth. Secondly, the grain-boundary film must be continuous. This is

the greatest drawback to the old cementite boundary hypothesis for, while some grain-boundary layers of cementite are observed in microsections, they are rare. There is, of course, the possibility that the films are too thin to be observed microscopically, but even here there is the objection that it would take an appreciable time to nucleate new cementite at the grain boundaries during the ageing process. No such induction interval is observed. Thus, the possibilities of a second phase at the grain boundaries must be precluded.

It is therefore necessary to seek a solution to this problem in terms of the carbon and nitrogen which is present, not as cementite or iron nitride, but held in solution in the matrix. Wert (1950) has recently shown, by internal friction methods, that the solubility of carbon in α -iron at room temperature is of the order of 10^{-7} weight per cent. A result of this order is to be expected from Hagg's rule. The ratio of the atomic diameters of iron and carbon is 0.63, whereas the limiting value, according to Hagg, is 0.59. Thus, carbon will exist preferentially in the α -iron lattice at places where the lattice is distorted, and large distorted areas occur at the grain boundaries. It seems possible that carbon will accumulate in these regions, forming areas of carbon-rich material which may have the necessary mechanical properties.

Evidence for grain-boundary segregation is apparently supplied by the etching characteristics of mild steel. If a specimen is so treated by any normal etchant, the grain boundaries come into relief as ridges, which are easily seen if a plastic replica of the surface is taken and examined under the electron microscope. The photographs of Trotter, McClean and Clews (1949) may be taken as typical examples. These ridges do not seem to be cementite, as there is no sharp boundary delineating the second phase. It would be interesting if it could be shown that these ridges are of carbon-rich material, of higher strength than the matrix, for films of this type would give rise to the yield point extension observed in mild steel.

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Reversible Effects in the Magnetization of Nickel

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ABSTRACT. An account is given of investigations on the temperature dependence of the contribution from reversible processes to the magnetization of annealed and strained nickel wires. This involves the measurement of reversible susceptibility κ_r over a range of field, and of temperature (-50°C. to $+200^\circ\text{C.}$), and a rapid automatic method of recording (κ_r, H) curves has been developed. The effect of temperature on κ_r is discussed in relation to Becker's strain theory and reasons for the discrepancies are suggested. It is shown that the contribution of reversible processes to the total change in magnetization, integrated over the hysteresis cycle, increases with temperature and strain, from 15% for an annealed specimen at -50°C. to 85% for a strained specimen (17.1 kg. mm^{-2}) at 150°C. The effect is not uniform over the whole cycle and in the region of the coercive field a considerable part of the change in magnetization must be attributed to irreversible processes.

§1. INTRODUCTION

IN a recent paper (Tebble and Corner 1950, to be referred to as I) an account is given of investigations on the reversible susceptibility κ_r of ferromagnetic materials at room temperature. Estimates were made, *inter alia*, of that part of the change in magnetization which takes place reversibly as the specimen is taken round a hysteresis cycle.

The method involves the measurements of κ_r at various points on the magnetization (I, H) curve. The area under the (κ_r, H) curve is given by

$$\int_{-H_m}^{+H_m} \kappa_r dH = \int_{-H_m}^{+H_m} \left(\frac{dI}{dH} \right)_r dH = \Delta I_r, \quad \dots\dots (1)$$

where H_m is the maximum applied field in each direction. This expression gives the *minimum* contribution of reversible processes to the total change in magnetization ΔI over the range of field considered. This method of estimating the reversible component of the change in magnetization makes possible the examination of the effect of temperature on the relative importance of the two groups of processes, reversible and irreversible. (The significance of the terms 'reversible' and 'irreversible' is discussed in I, §§1 and 2.) For this purpose an automatic method has been developed whereby (κ_r, H) curves can be recorded rapidly while the temperature of the specimen (in the range -50°C. to $+200^\circ\text{C.}$) is held constant and measured accurately. It should be mentioned that a direct

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